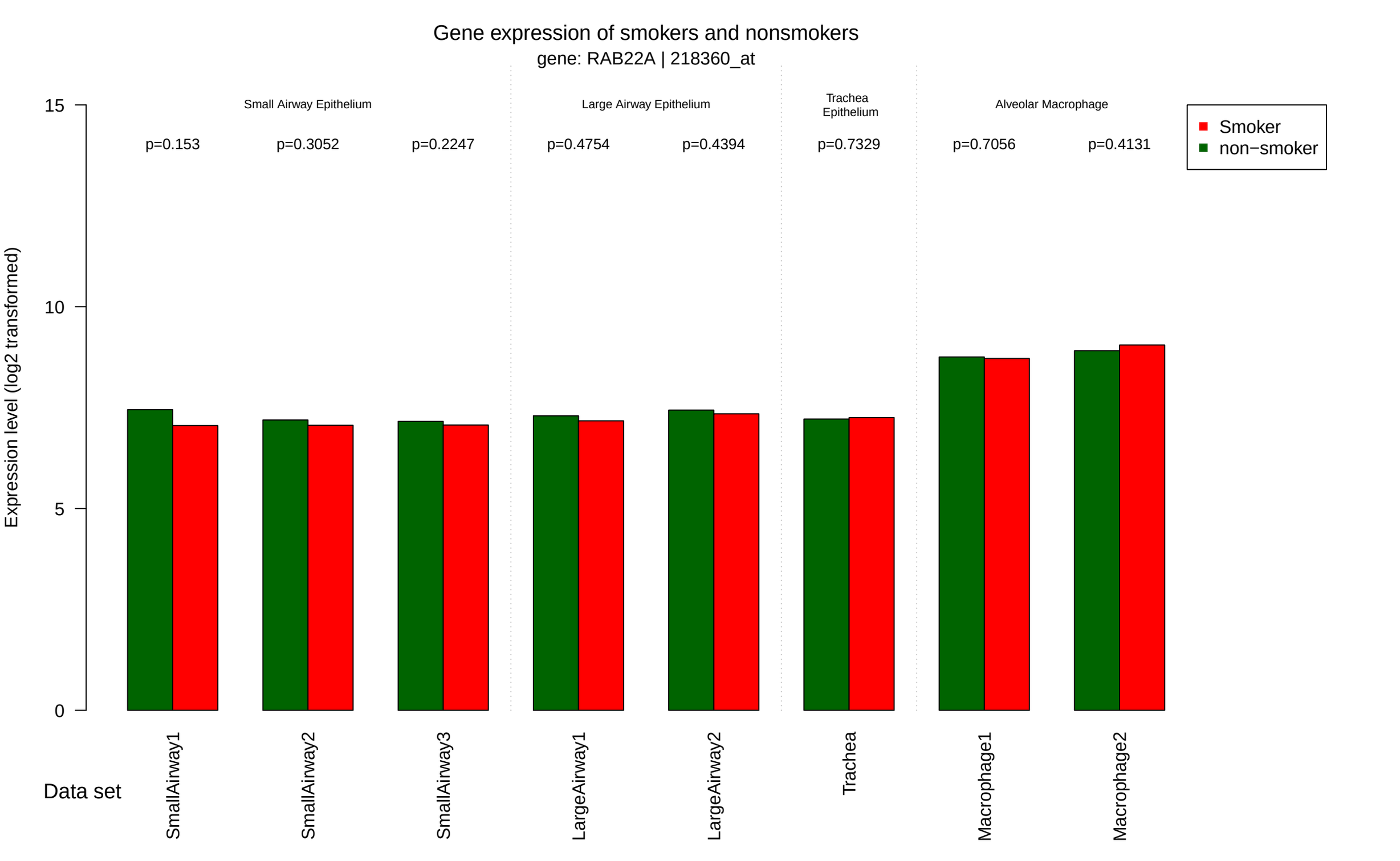
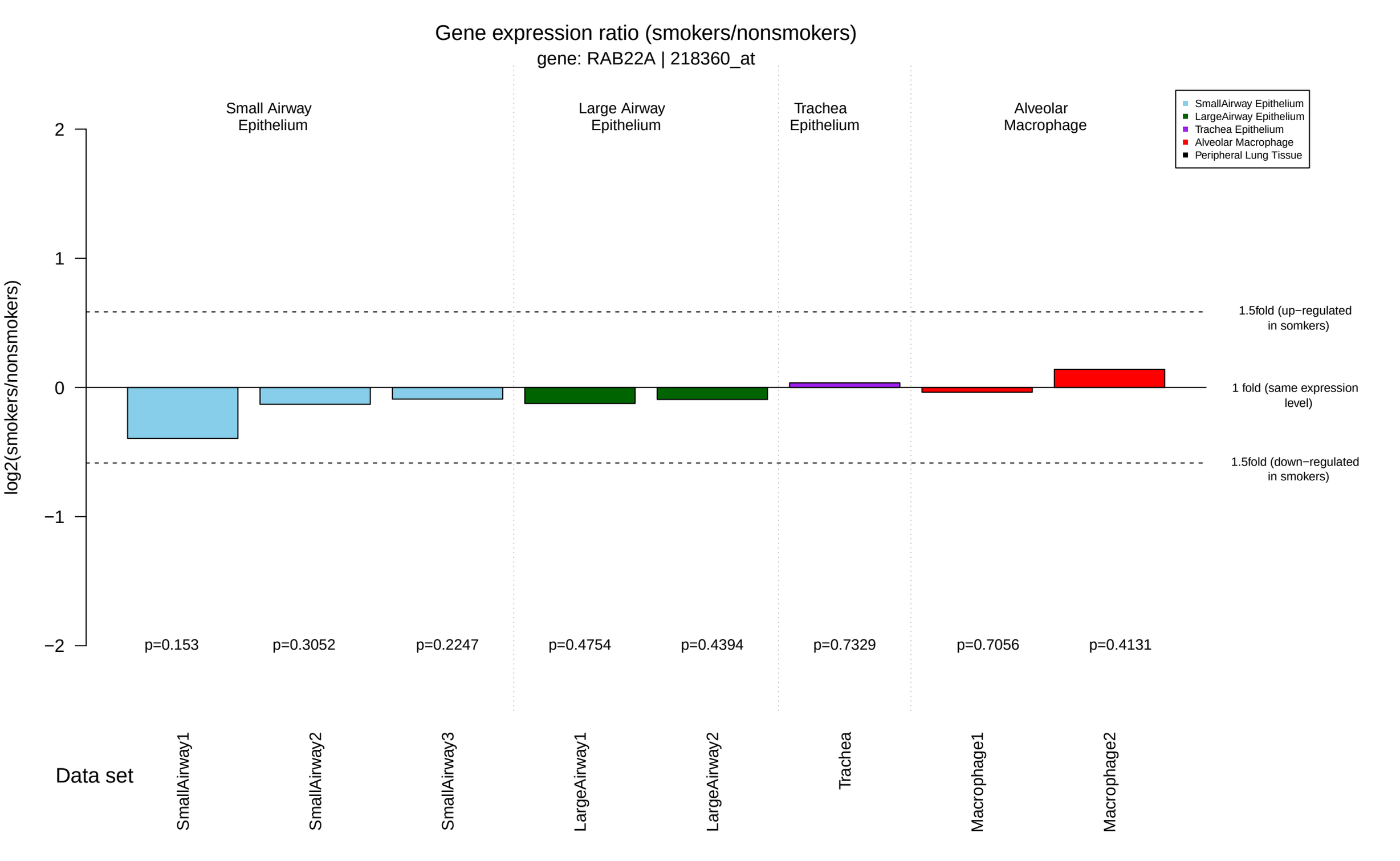
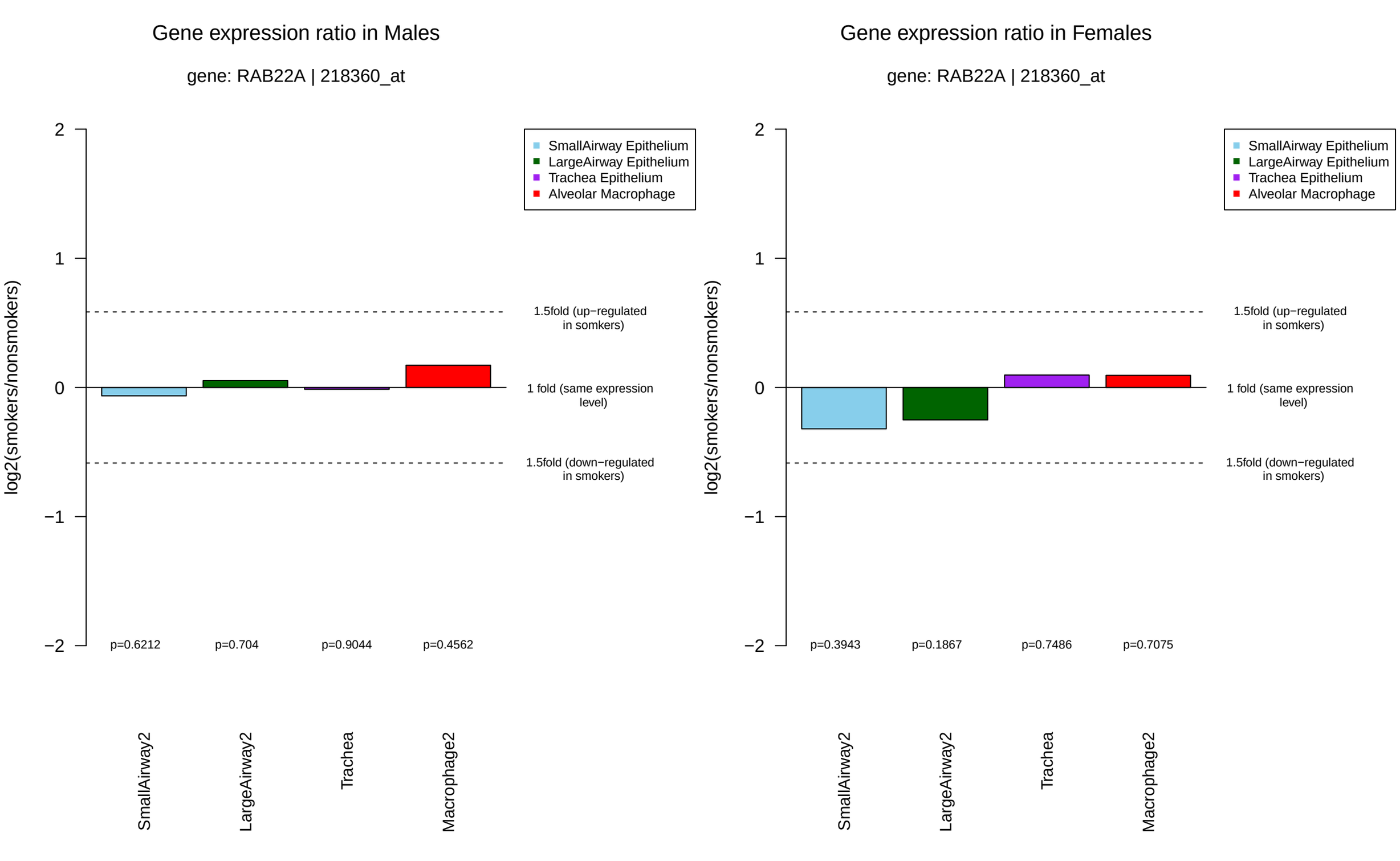
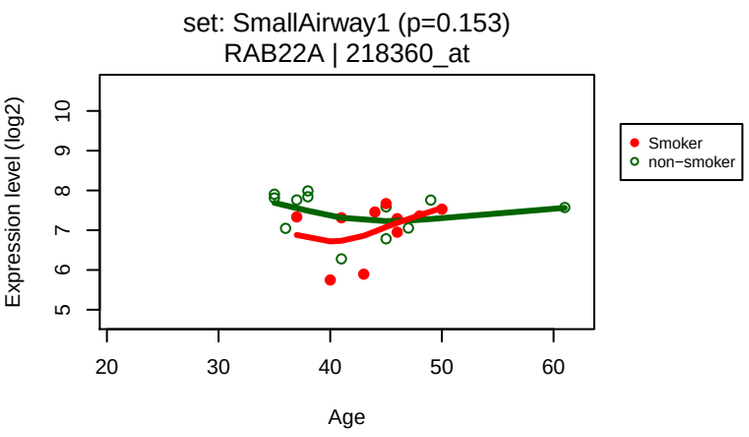
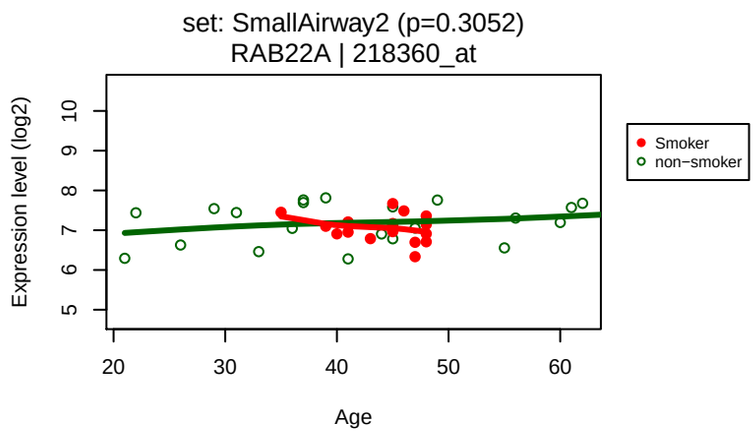
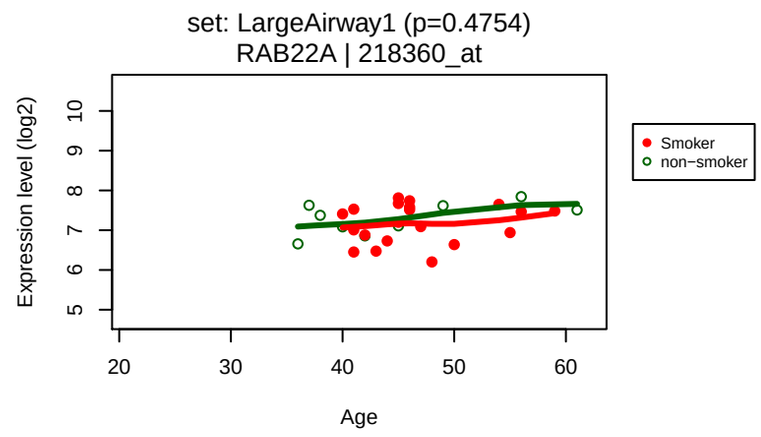

Gene expression of smokers and nonsmokers
gene: RAB22A | 218360_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.153
p=0.3052
p=0.2247
p=0.4754
p=0.4394
p=0.7329
p=0.7056
p=0.4131
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

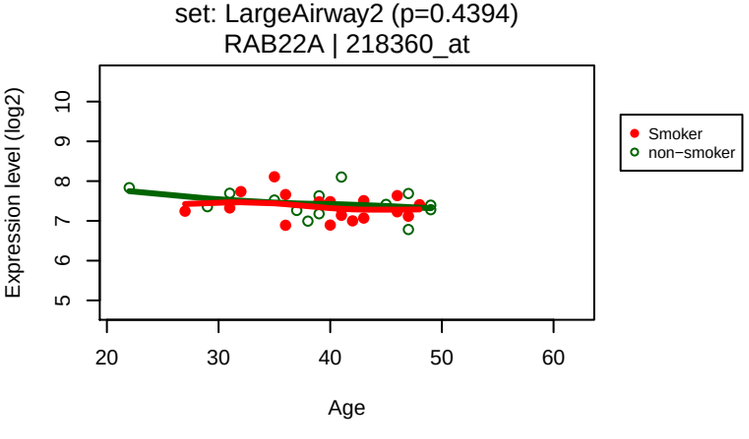

Gene expression ratio (smokers/nonsmokers)
gene: RAB22A | 218360_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.153
p=0.3052
p=0.2247
p=0.4754
p=0.4394
p=0.7329
p=0.7056
p=0.4131
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

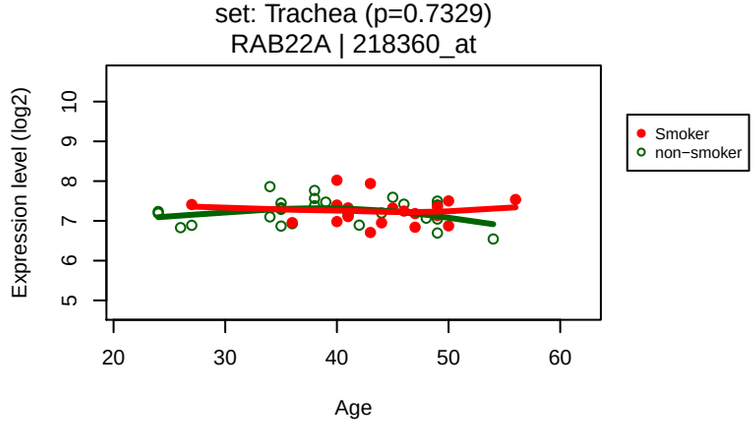

Gene expression ratio in Males
gene: RAB22A | 218360_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6212
p=0.704
p=0.9044
p=0.4562
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB22A | 218360_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3943
p=0.1867
p=0.7486
p=0.7075
SmallAirway2
LargeAirway2
Trachea
Macrophage2

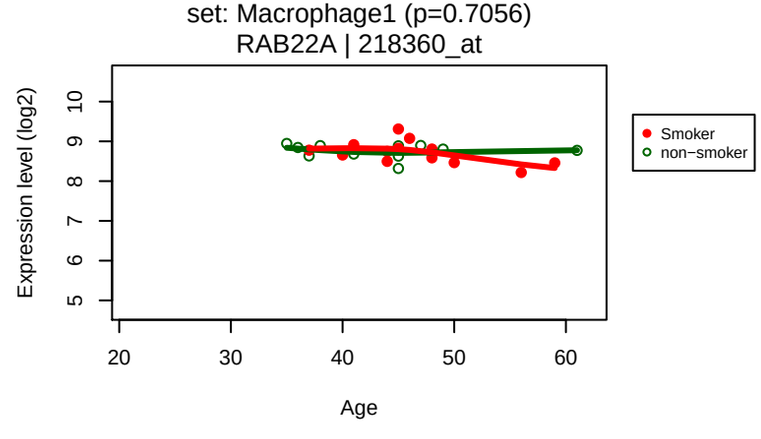

set: SmallAirway1 (p=0.153)
RAB22A | 218360_at
Expression level (log2)
Age
Smoker
non−smoker

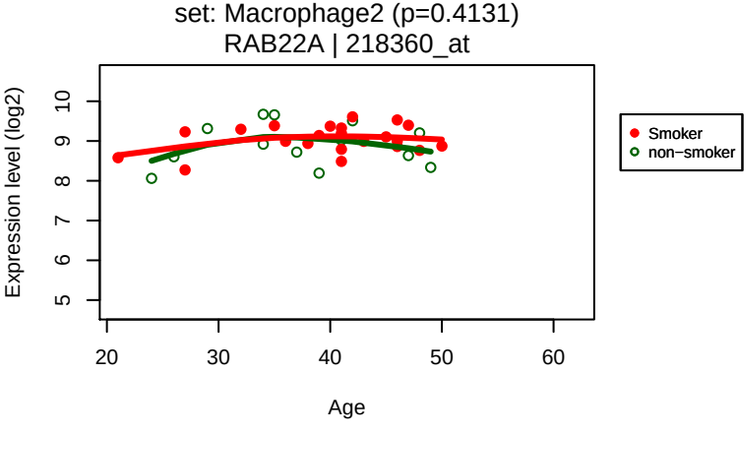

set: SmallAirway2 (p=0.3052)
RAB22A | 218360_at
Expression level (log2)
Age
Smoker
non−smoker

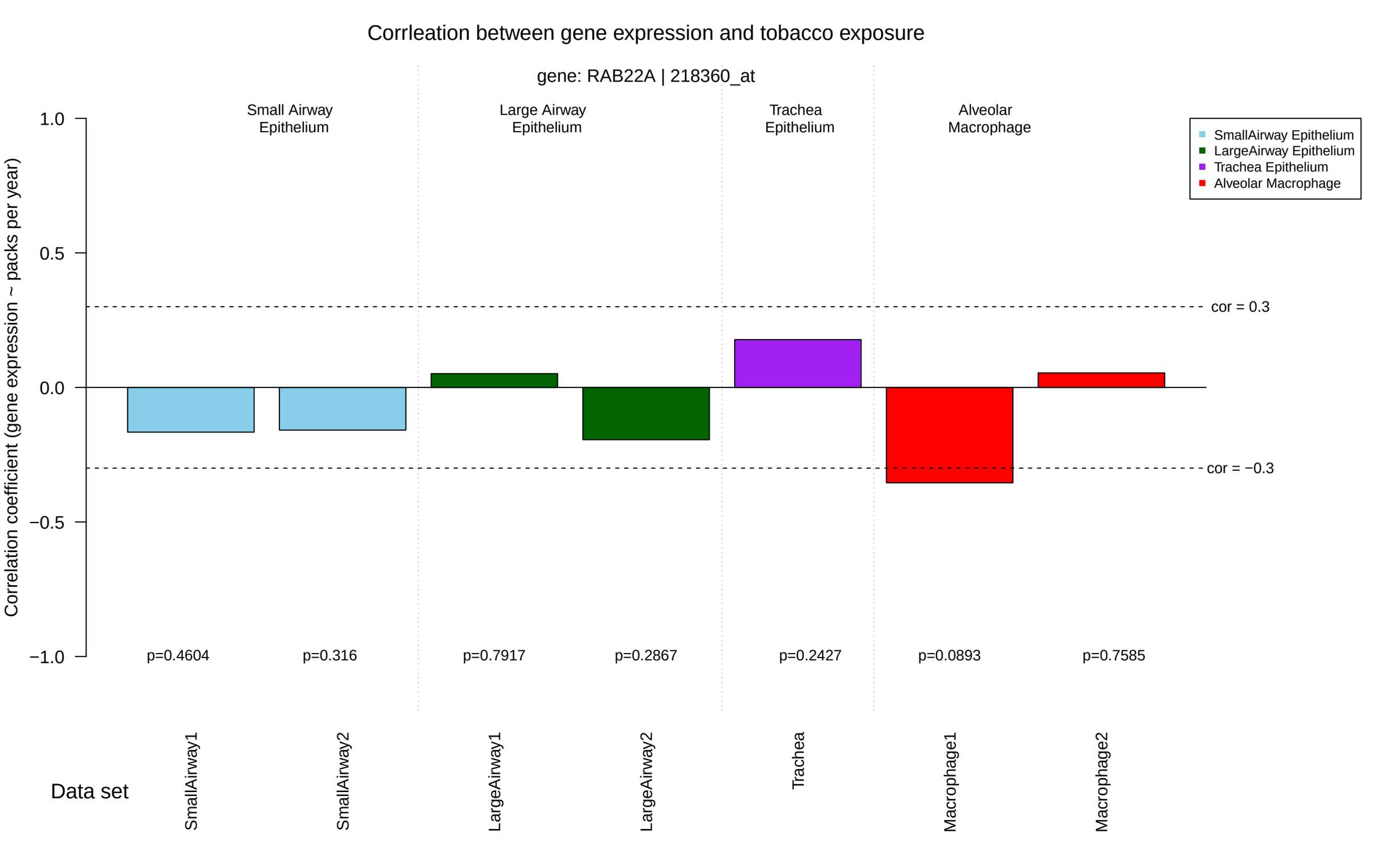

set: LargeAirway1 (p=0.4754)
RAB22A | 218360_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4394)
RAB22A | 218360_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7329)
RAB22A | 218360_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7056)
RAB22A | 218360_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4131)
RAB22A | 218360_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB22A | 218360_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4604
p=0.316
p=0.7917
p=0.2867
p=0.2427
p=0.0893
p=0.7585
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2